SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I S. Roshan (192126011) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	2
Concept Mapping Structure	25%	2	2	2
Linking & Relationships	20%	1	1	2
Application & Analysis	20%	1	1	1
Explanation & Clarity	15%	1	1	1

Total Marks 22 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

Serial no: 29

SIMATS ENGINEERING

Assessment Tool - 2

Co - 1

Concept Mapping

22/30

Course Name : Computer networks

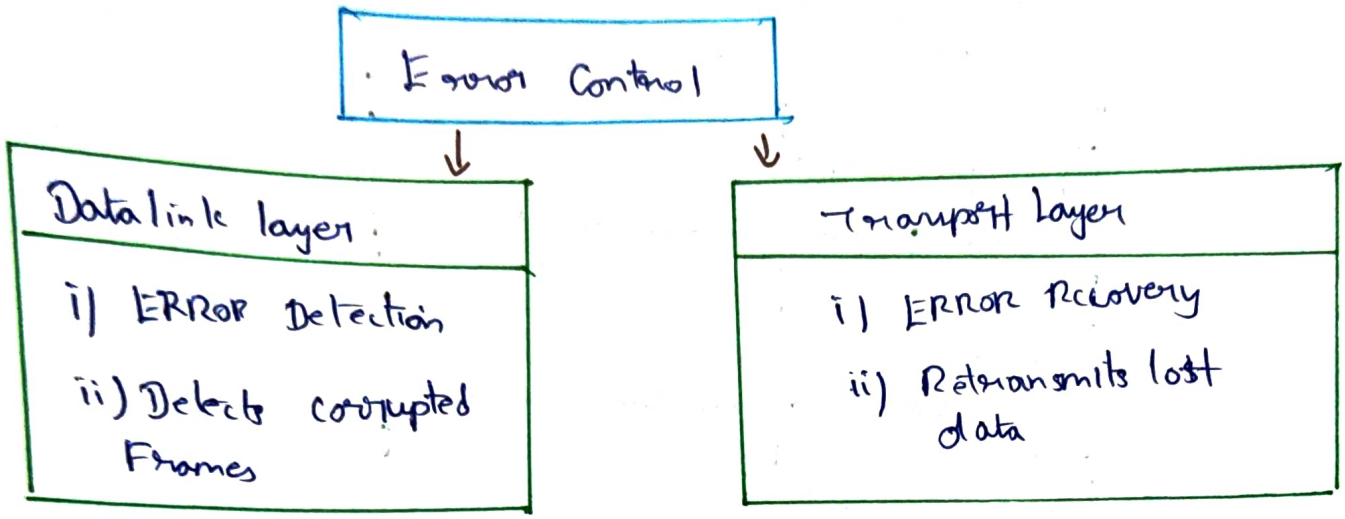
Course Code : CSA0709

Name : S. Rathan

Regno : 191126001H1

ERROR CONTROL IN DATA TRANSMISSION

Concept map :



Analysis :

1) The Data Link detects errors in Frames using methods such as CRC

2) The Transport layer, using TCP, detects missing or incorrectly received data and request retransmission when necessary

3) Both layers work together to ensure data reaches the destination correctly.

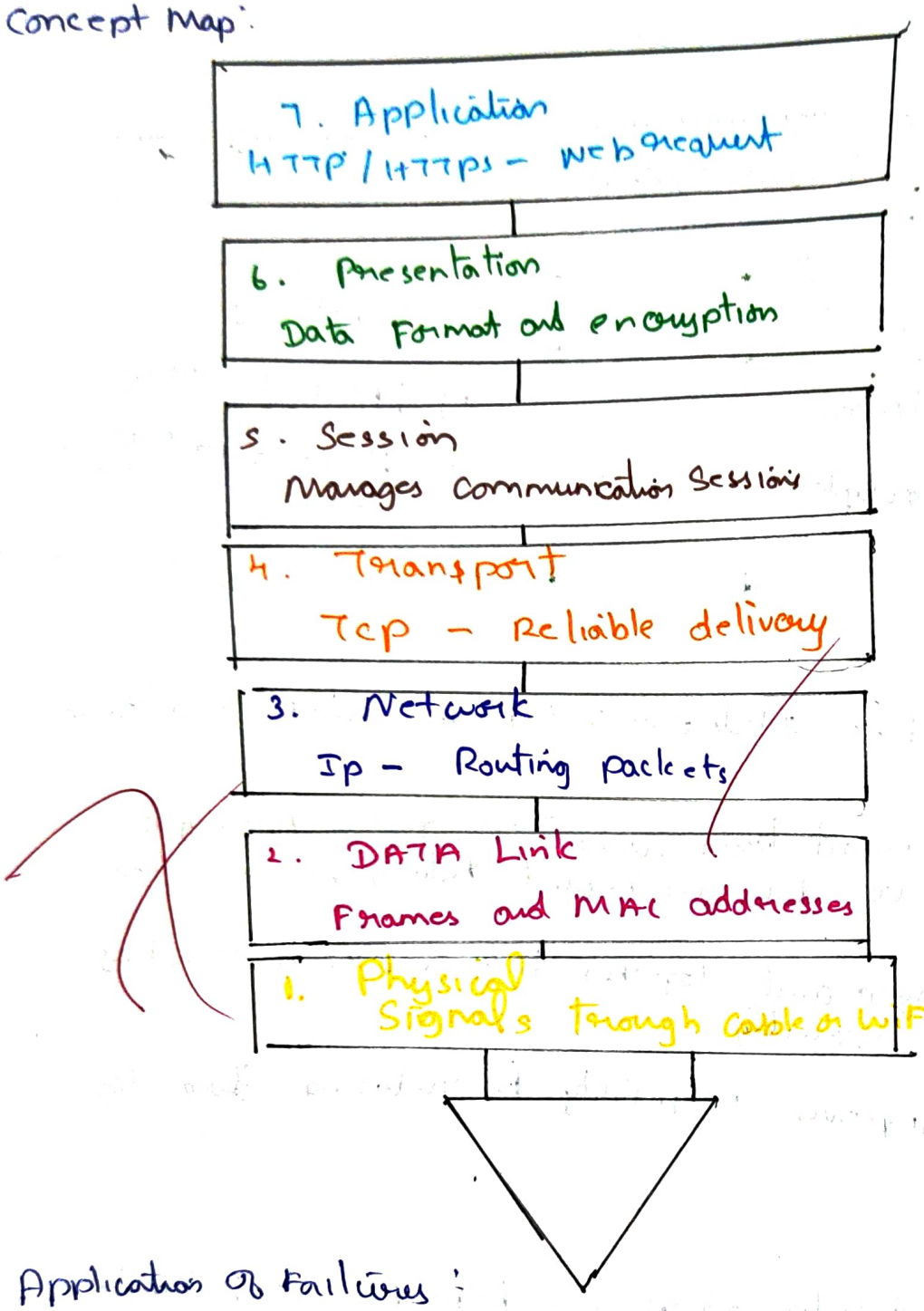
4) This improves reliability by reducing data loss and corruption.

Conclusion :

ERROR detection and recovery helps provide reliable data transmission.

5) Concept Map For Web Browsing using OSI layers.

Concept Map:



Application of Failures:

* Application layer: Incorrect URL or browser errors prevents access to a website.
* Presentation Layer: Encryption or data-format problems may prevent the page from loading correctly.
* Session layer: Session Failure may interrupt communication.
* Transport layer: Tcp problems can cause delays

reaching this Server.

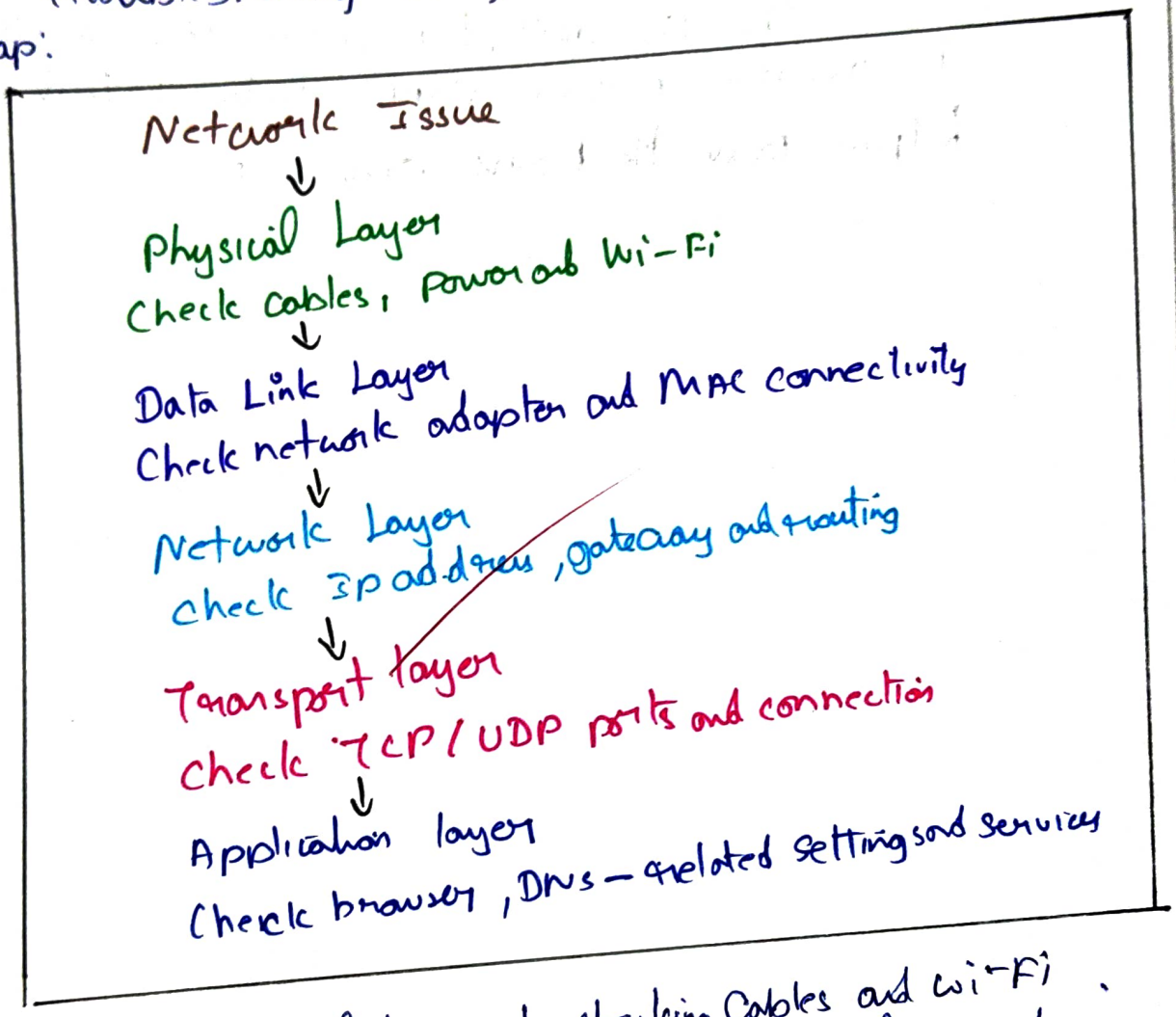
Data link layer : WiFi or MAC - address problem disrupt local network communication.

Physical layer : A damaged cable or weak Wi-Fi signal causes connection failure.

Conclusion : The OSI model helps identify which layer is responsible for a web - browsing problem.

Network Troubleshooting Using the OSI Model.

Concept Map:



Analysis:

1) Start with Physical Layer by checking Cables and Wi-Fi.
2) Check the Data link Layer for adapter and Local connecting problems
3) Verify the ip address, default gateway and routing at the Network layer.

4) Check the ports and TCP/UDP connections at Transport layer
5) Finally, check the browser and application services.

How the concept map helps:

* It provide a systematic trouble shooting order.
* It helps isolate the source of the problem
* It avoid unnecessary checks of unrelated components.
* It makes network Fault identification faster and easier.

Conclusion:

The OSI concept map simplifies trouble shooting by dividing network communication into layers and helping locate the Faulty component.